Horizon-Dependent Bitcoin Return Forecasting: Evidence from Adaptive Forecast Combination and Rigorous Out-of-Sample Evaluation

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Abstract

Bitcoin forecasting performance may depend strongly on the horizon at which predictions are required. This study evaluates that possibility using eight direct cumulative-return horizons from 15 days to 36 months and a heterogeneous set of benchmark, statistical, neural, hybrid, and adaptive-combination forecasts. An expanding walk-forward design preserves the information boundary at every forecast origin, while return RMSE and MAE are complemented by overlap-robust Diebold–Mariano tests, a block-bootstrap model-set diagnostic, directional measures, and robustness checks. The results show pronounced horizon dependence. Historical Median is the lowest-RMSE specification at 15 days, whereas the Adaptive Ensemble becomes the empirical RMSE leader from one month onward. Its improvement over the best non-ensemble benchmark is 9.57% at two months and rises above 60% at 24 and 36 months; the corresponding advantage is statistically distinguishable from the best non-ensemble benchmark from two months onward, but not at one month. Model-set uncertainty nevertheless remains substantial. Directional accuracy also requires qualification: at 36 months, 99.68% of realized returns are positive, while the Adaptive Ensemble reaches 99.81% accuracy, only 0.13 percentage points above an Always-Up rule. The evidence therefore supports horizon-dependent forecast combination, not claims of near-perfect conditional Bitcoin predictability.

Keywords: Bitcoin; cryptocurrency forecasting; return forecasting; forecast combination; adaptive ensemble; LSTM; walk-forward evaluation; Diebold–Mariano test; model uncertainty; directional accuracy.

1 Introduction

Bitcoin provides an unusually demanding environment for empirical forecasting. It trades continuously, is not issued by a conventional central bank or corporation, and has developed within a market structure in which speculative demand, technological adoption, network activity, liquidity, and macro-financial conditions interact. Its return distribution is also time-varying. These characteristics make Bitcoin useful for studying a question that is broader than whether a particular machine-learning algorithm can obtain a low prediction error: *does the relative usefulness of forecasting models depend on the horizon at which the forecast is required?*

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The empirical literature offers no universal model ranking. Early work examined Bitcoin price formation through monetary, market, user, and production-cost mechanisms (Ciaian et al., 2016; Hayes, 2017), while frequency-domain evidence documented variation in the relationship between Bitcoin and potential drivers (Kristoufek, 2015). Network and blockchain variables have also received attention. Koutmos (2018) studied Bitcoin returns and transaction activity, Aalborg et al. (2019) examined determinants of Bitcoin returns, volatility, and trading volume, and Guo et al. (2021) showed that blockchain transaction information can contribute to cryptocurrency price forecasting. These studies motivate richer information sets but do not establish that a given predictor or model should remain useful across all forecast horizons.

The forecasting literature has increasingly moved from classical econometric methods toward machine learning and deep learning. Comparative evidence remains heterogeneous. Berger and Koubová (2024) found that machine-learning methods can outperform econometric benchmarks for Bitcoin returns, while also reporting that additional LSTM complexity does not necessarily improve daily forecasts. AlMadany et al. (2024) compared classical and deep-learning models across major cryptocurrencies and found modest advantages for hybrid volatility–LSTM specifications. Bouteska et al. (2024) reported that the relative performance of ensemble and deep-learning methods varies across cryptocurrencies and model classes. More recent studies extend the information set toward textual, blockchain, and high-dimensional factor data (Gurgul et al., 2025; Bagheri and Giudici, 2025; Li et al., 2026). The growing methodological diversity makes a single model ranking increasingly difficult to interpret without explicit attention to target definition and forecast horizon.

A second issue concerns forecast combination. Forecast combinations have a long history in econometrics. The basic argument is that different models can contain complementary information and need not have perfectly correlated forecast errors (Bates and Granger, 1969; Timmermann, 2006). Combination can therefore improve prediction even when no component is uniformly superior. At the same time, equal weighting may give excessive influence to unstable forecasts, while data-dependent weighting can introduce look-ahead bias if the weighting information is not strictly out of sample. These concerns are especially relevant in a market in which predictive relationships may evolve.

This study addresses these issues through a horizon-centered forecasting framework. Rather than treating Bitcoin forecasting as a single task, we evaluate direct cumulative returns at eight horizons: 15 days, 1, 2, 3, 6, 12, 24, and 36 months. The candidate set includes simple benchmarks, ARIMA, OLS, GLS, LSTM, a volatility-sensitive LSTM, a GLS-LSTM hybrid, and an adaptive ensemble. The ensemble uses only previously observed out-of-sample forecast errors to update inverse-error weights. The design therefore permits a direct empirical assessment of whether the value of combination changes with the forecast horizon.

The study makes five contributions. First, it places forecast horizon at the center of model comparison rather than treating horizon as a secondary robustness dimension. Second, it evaluates adaptive combination against heterogeneous statistical and neural forecasts under an expanding walk-forward design. Third, it combines magnitude-based forecasting metrics with HAC-based pairwise inference and a conservative model-set diagnostic. Fourth, it explicitly separates directional performance from the unconditional distribution of realized return signs.

This distinction becomes crucial at very long horizons, where the positive-return base rate becomes highly asymmetric. Fifth, the study provides an auditable computational design in which forecast origins, retraining frequency, model specifications, and forecast records are retained for independent reconstruction.

The main finding is not that one model universally dominates Bitcoin forecasting. Instead, the evidence indicates a pronounced horizon effect. Historical Median is strongest at 15 days, whereas adaptive combination becomes the lowest-RMSE specification from one month onward. The gain is not statistically distinguishable at one month but becomes statistically distinguishable from two months onward. At the same time, the model-set diagnostic does not isolate a unique superior model. The appropriate interpretation is therefore that adaptive combination is a useful horizon-dependent forecasting strategy under the present evaluation design, not that it establishes a universal law of Bitcoin predictability.

2 Research Questions and Hypotheses

The empirical analysis is organized around four research questions:

RQ1: Does the relative forecasting performance of statistical, neural, hybrid, benchmark, and ensemble models vary systematically with the Bitcoin forecast horizon?

RQ2: Does an adaptive combination of heterogeneous forecasts improve out-of-sample return prediction relative to individual non-ensemble models?

RQ3: Are the observed improvements statistically distinguishable after accounting for serial dependence induced by overlapping multi-horizon targets?

RQ4: To what extent do apparently high directional-accuracy rates at long horizons reflect genuine conditional forecasting information rather than the unconditional sign distribution of realized returns?

The hypotheses are stated before the empirical results and are intended to be falsifiable.

H1 (Horizon dependence). The relative ranking of forecasting models varies across forecast horizons; no single model is expected to dominate at all horizons.

H2 (Adaptive combination). The Adaptive Ensemble achieves lower out-of-sample return RMSE than the best non-ensemble specification at medium and long horizons.

H3 (Statistical distinguishability). The predictive-accuracy advantage of the Adaptive Ensemble over the best non-ensemble specification is expected to emerge more clearly at medium and long horizons than at the shortest horizons, as assessed using overlap-robust Diebold–Mariano tests.

H4 (Model uncertainty). Point-error rankings need not imply that one model can be statistically isolated from all alternatives under a model-set procedure.

H5 (Directional base-rate qualification). Very high directional accuracy at long horizons is partly attributable to the unconditional positive-return base rate and should therefore not automatically be interpreted as evidence of near-perfect conditional predictability.

3 Literature Review

3.1 Bitcoin return predictability

The early empirical literature on Bitcoin focused primarily on price formation and market efficiency. Ciaian et al. (2016) examined market forces and Bitcoin-specific determinants of price formation, while Hayes (2017) developed a production-cost perspective. Kristoufek (2015) documented time-frequency variation in the relationship between Bitcoin and potential explanatory variables. Together, these studies emphasize that Bitcoin valuation is shaped by mechanisms whose relevance can vary over time.

Network and blockchain variables provide another source of information. Koutmos (2018) examined the relationship between Bitcoin returns and transaction activity. Aalborg et al. (2019) studied returns, volatility, and trading volume using market and blockchain-related variables, while Guo et al. (2021) showed that underlying blockchain transactions can contribute to cryptocurrency price forecasting. Such findings motivate the inclusion of on-chain variables in the present forecasting information set, but they do not imply that predictive relationships remain stable across horizons.

Evidence on market efficiency is mixed. Urquhart (2016) and Nadarajah and Chu (2017) reached different conclusions concerning weak-form inefficiency, while Aslan and Sensoy (2020) documented frequency-dependent efficiency patterns. More recent work has continued to show that Bitcoin return predictability depends on the information set, sample period, and model class. Rudkin et al. (2025), for example, show that directional forecast performance can depend on the historical return trajectory. This is closely related to the present study's emphasis on horizon-specific evaluation and caution in interpreting directional accuracy.

3.2 Classical statistical and machine-learning forecasting

Classical statistical models remain important because they provide transparent benchmarks and can perform well when the signal-to-noise ratio is low. Machine learning can represent nonlinear interactions, but greater complexity does not automatically imply better out-of-sample forecasts. Berger and Koubová (2024) provide a direct Bitcoin comparison between econometric time-series and machine-learning approaches and find that model complexity does not uniformly translate into better daily forecasts.

Recent comparative evidence reinforces the absence of a universal winner. AlMadany et al. (2024) compare ARMA/GARCH-family models and LSTM approaches across major cryptocurrencies, while Bouteska et al. (2024) compare ensemble and deep-learning methods across several digital assets. Gurgul et al. (2025) integrate financial, blockchain, and social-media information with deep learning and NLP, and Bagheri and Giudici (2025) compare accuracy, security, explainability, and robustness across several machine-learning architectures. Li et al. (2026) use high-dimensional cryptocurrency and macroeconomic factors and find strong performance from tree-based models relative to neural and linear benchmarks. These results highlight the importance of treating model comparisons as conditional on the forecasting problem rather than as a permanent hierarchy of algorithms.

3.3 Forecast combination

Forecast combination is motivated by model uncertainty. Bates and Granger (1969) demonstrated that combinations can reduce forecast error when component forecasts contain complementary information. Timmermann (2006) subsequently reviewed the theoretical and empirical determinants of combination gains, including forecast-error correlation, model instability, and estimation uncertainty.

The present study does not assume that inverse-error weighting is optimal. Instead, it uses a transparent causal rule in which each component receives a weight proportional to the inverse of its recent mean absolute out-of-sample error. The purpose is to test whether recent predictive performance can be exploited without using information from the current or future realization. This distinction is important: the ensemble is a transparent adaptive heuristic, not a statistically optimal superlearner.

3.4 Forecast evaluation and model uncertainty

Point-error rankings alone do not establish predictive superiority. The Diebold–Mariano framework provides a formal comparison of predictive accuracy (Diebold and Mariano, 1995), while Harvey et al. (1997) provide a small-sample refinement. Multi-horizon targets overlap, so independence-based standard errors are inappropriate. The present analysis therefore uses HAC estimation for loss differentials, with the dependence horizon linked to the forecast horizon.

Model-selection uncertainty is a separate issue. The Model Confidence Set framework of Hansen et al. (2011) is designed to identify a set of models that cannot be statistically distinguished from the best-performing model. The present implementation uses a block-bootstrap Hansen-style model-set diagnostic rather than a literal reproduction of every step of the canonical finite-sample MCS algorithm. It is therefore interpreted conservatively as evidence about model-set uncertainty rather than as a formal MCS certification.

3.5 Positioning of the present study

The contribution of this study lies at the intersection of these strands. It does not introduce a new neural architecture. Instead, it asks whether model performance and the usefulness of forecast combination vary with horizon when the evaluation is performed consistently. The question is particularly relevant for Bitcoin because cumulative-return targets become statistically different objects as the horizon increases and because directional metrics can become misleading when the unconditional sign distribution becomes highly asymmetric.

4 Data and Forecasting Design

4.1 Sample and target

The frozen modeling dataset contains 4,761 daily observations spanning 14 May 2011 to 25 May 2024. The analysis uses a direct cumulative log-return target. Let P_t denote the Bitcoin closing price at day t. For horizon h,

$$R_{t,h} = 100 \log \left(\frac{P_{t+h}}{P_t} \right). \tag{1}$$

The horizons are

$$h \in \{15, 30, 60, 90, 180, 365, 730, 1095\}. \tag{2}$$

Thus, a 36-month model directly forecasts the 36-month cumulative return. It does not recursively iterate a one-day price forecast.

The direct-target construction determines the information boundary. At forecast origin t, the latest target that can be observed is $R_{t-h,h}$, because that target ends at t. Consequently, with an initial training requirement of 1,000 target observations, the first valid origin is

$$t_0 = 1000 + h - 1. \tag{3}$$

The number of valid forecast origins follows mechanically as

$$N_h = N - (1000 + h - 1) - h, \tag{4}$$

where $N = 4761$. For example, at $h = 1095$ this gives $4761 - (1000 + 1095 - 1) - 1095 = 1572$, matching the 36-month evaluation count in Table 1. This explicit accounting is important because the effective evaluation sample necessarily decreases as the horizon increases.

4.2 Predictors and information boundary

The principal information set contains active addresses, transaction count, mining difficulty, network hashrate, block size, transaction value, average transaction value, Bitcoin price, daily log return, lagged return, seven-day rolling volatility, thirty-day rolling volatility, day of week, month, and year.

Highly skewed positive variables are transformed using $\log(1 + x)$ before standardization. All scaling parameters are estimated using only the training information available at each origin. The volatility-sensitive neural specification uses a reduced feature set containing active addresses, difficulty, hashrate, price, return, lagged return, and the two rolling-volatility measures.

4.3 Walk-forward evaluation

The evaluation follows an expanding-window design. The initial training sample contains 1,000 usable target observations. Models are retrained every 25 forecast origins. This frequency is treated as a prespecified computational design choice rather than an empirically optimized hyperparameter.

Table 1: Out-of-sample forecast origins by horizon.

Horizon	Days	Valid origins
15D	15	3,732
1M	30	3,702
2M	60	3,642
3M	90	3,582
6M	180	3,402
12M	365	3,032
24M	730	2,302
36M	1,095	1,572

The resulting forecast panel contains 324,558 model-origin records across all horizons and 13 forecasting specifications. The 13 specifications are Zero Return, Historical Mean, Random Walk with Drift, Historical Median, Majority Direction, Persistence Direction, ARIMA, OLS, GLS, LSTM, VS-LSTM, GLS-LSTM, and Adaptive Ensemble. Direction-only benchmarks are retained for directional evaluation but are not treated as magnitude-forecast components.

5 Forecasting Models

5.1 Benchmark models

The Zero Return benchmark predicts a zero cumulative return. Historical Mean predicts the mean of previously observed direct targets, while Historical Median predicts the corresponding historical median. Random Walk with Drift produces the same magnitude forecast as Historical Mean under the present direct-return formulation and is therefore not included separately in the Adaptive Ensemble. Majority Direction and Persistence Direction are evaluated separately because they are directional heuristics and do not provide independent magnitude forecasts suitable for the return-RMSE ensemble.

5.2 ARIMA

An ARIMA$(1, 0, 1)$ model is estimated directly on the sequence of horizon-specific cumulative returns $R_{t,h}$. At forecast origin t, only target observations whose future endpoints are already observable are available for estimation. The model then produces the required multi-step forecast of the direct-return sequence, with the final forecast corresponding to the cumulative return target at the current forecast origin. This is therefore direct cumulative-return forecasting, not recursive price forecasting.

5.3 OLS and GLS

The linear models use the information set described above. OLS provides a transparent linear benchmark. GLS is estimated with a first-order autoregressive error structure using an iterative feasible procedure. The purpose of including GLS is to provide a statistically structured linear comparator rather than to assume that a particular error process is universally correct.

5.4 LSTM

The LSTM receives the most recent 30 observations of the modeling features. The network contains 32 LSTM units followed by dropout of 0.15, a 16-unit ReLU dense layer, and a single linear output. The Adam optimizer is used with a learning rate of 10^{-3} and mean squared error loss. Training is limited to 40 epochs, with early stopping after six epochs without improvement. A chronological validation segment comprising the final 10% of the training observations is used; observations are not shuffled.

5.5 Volatility-sensitive LSTM and hybrid model

The volatility-sensitive LSTM uses the reduced feature set described in Section 4. Its role is diagnostic: it tests whether a more focused information set improves forecasting relative to the broader feature specification. The GLS-LSTM hybrid combines the GLS-based statistical representation with the recurrent architecture and provides a heterogeneous component that may capture linear dependence and nonlinear temporal structure jointly.

5.6 Adaptive Ensemble

The Adaptive Ensemble combines nine magnitude-forecasting components: Zero Return, Historical Mean, Historical Median, ARIMA, OLS, GLS, LSTM, VS-LSTM, and GLS-LSTM.

Random Walk with Drift is excluded because it is identical to Historical Mean under the direct-return formulation. Majority Direction and Persistence Direction are not combined with magnitude forecasts because they are direction-only rules and do not supply an independent return-magnitude prediction.

Let $e_{m,t}$ denote the previously realized forecast error of model m. The recent error scale is

$$MAE^{(20)}_{m,t} = \frac{1}{20}\sum_{j=1}^{20} |e_{m,t-j}|. \tag{5}$$

The raw weight is

$$\widetilde{w}_{m,t} = \frac{1}{\max(MAE^{(20)}_{m,t}, \epsilon)}, \tag{6}$$

and the normalized weight is

$$w_{m,t} = \frac{\widetilde{w}_{m,t}}{\sum_j \widetilde{w}_{j,t}}. \tag{7}$$

The ensemble forecast is

$$\widehat{R}_{t,h}^{AE} = \sum_{m} w_{m,t}\widehat{R}_{m,t,h}. \tag{8}$$

Only errors available strictly before the current forecast are used. Consequently, the current realization cannot affect its own ensemble weight. When a fitted component is unavailable, it is excluded rather than assigned an artificial zero forecast.

The weighting rule is intentionally described as *adaptive inverse-error weighting*, not as an optimal or statistically efficient combination rule. The first 20 eligible forecasts use equal weights because no model has yet accumulated the required error history.

6 Evaluation Measures and Statistical Inference

The primary metric is return RMSE,

$$RMSE_h = \sqrt{\frac{1}{N_h}\sum_{t=1}^{N_h}(\widehat{R}_{t,h} - R_{t,h})^2}, \tag{9}$$

with MAE as a secondary magnitude metric,

$$MAE_h = \frac{1}{N_h}\sum_{t=1}^{N_h}|\widehat{R}_{t,h} - R_{t,h}|. \tag{10}$$

All RMSE and MAE values are expressed in percentage-return units because the target is $100\log(P_{t+h}/P_t)$.

Directional accuracy is

$$DA_h = \frac{1}{N_h^*}\sum_{t} I\{\text{sign}(\widehat{R}_{t,h}) = \text{sign}(R_{t,h})\}, \tag{11}$$

where zero predicted or realized returns are excluded from the directional comparison. Balanced accuracy and the Matthews correlation coefficient (MCC) are also reported to reduce sensitivity to class imbalance.

For pairwise forecast comparison, the loss differential between models A and B is

$$d_t = L(e_{A,t}) - L(e_{B,t}), \tag{12}$$

where the primary loss is squared error. The Diebold–Mariano statistic is computed using a HAC estimate of the long-run variance. The bandwidth is linked to the forecast horizon so that dependence induced by overlapping targets is not treated as independence. Comparisons with the best non-ensemble model use a one-sided alternative in favor of lower expected loss for the Adaptive Ensemble.

6.1 Model-set diagnostic

A block-bootstrap Hansen-style range diagnostic is used to assess whether observed cross-model differences are sufficiently large to isolate a statistically distinguishable superior subset. The diagnostic is evaluated at $\alpha = 0.05$, 0.10, and 0.15. It is deliberately interpreted conservatively: because it is an approximation rather than a literal reproduction of the full Hansen–Lunde–Nason implementation, its output is treated as evidence about model-set uncertainty rather than as a canonical MCS certification (Hansen et al., 2011).

6.2 Directional base-rate diagnostic

For each horizon, the unconditional fraction of positive realized returns is calculated. Always-Up accuracy is then compared with the Adaptive Ensemble's directional accuracy. The exact McNemar test is used to compare paired directional classifications. A non-rejection of the McNemar null is not interpreted as evidence of equivalence; it only indicates that the directional classifications are not statistically distinguishable under that test.

6.3 Robustness and stability

Recent-sample performance is calculated on the latter half of the evaluation observations. Non-overlapping target observations are also evaluated as a diagnostic robustness exercise. The latter is not treated as a primary test because the effective sample size becomes extremely small at long horizons.

CUSUM diagnostics examine temporal stability of the forecast-error mean. ADF and KPSS tests are used descriptively to assess how the statistical properties of the direct-return target change with horizon. Given overlapping cumulative returns, neither diagnostic is interpreted as a standalone proof of a particular data-generating process.

7 Results

7.1 Overall forecast accuracy

The complete return-RMSE comparison is reported in Table 2 and summarized visually in Figure 1. The model ranking changes materially with the horizon.

Table 2: Return RMSE for the magnitude-forecasting specifications across horizons. Lower values indicate better forecasts. Values are percentage-return units.

Model	15D	1M	2M	3M	6M	12M	24M	36M
Adaptive Ensemble	**14.94**	**21.00**	**29.62**	**34.96**	**46.88**	**52.38**	**42.08**	**37.12**
Historical Median	**14.61**	21.67	32.75	42.58	67.09	100.51	129.84	177.96
Zero Return	14.67	21.93	33.09	42.72	66.95	108.84	144.09	142.35
Historical Mean	14.76	22.05	33.11	43.16	67.19	100.82	142.11	155.09
LSTM	14.92	22.00	34.36	44.72	69.44	101.07	119.02	97.28
VS-LSTM	15.01	22.89	33.86	43.57	69.64	105.22	130.34	114.91
ARIMA	15.13	23.37	34.78	44.49	70.87	120.30	173.03	147.68
OLS	15.84	25.50	44.71	65.26	96.99	145.09	129.66	176.90
GLS-LSTM	71.68	918.31	1690.03	1815.00	91.42	140.92	181.12	170.33
GLS	319.44	1427.52	2147.64	2125.70	133.91	133.88	164.51	359.78

At 15 days, Historical Median has the lowest RMSE (14.61), followed by Zero Return, Historical Mean, LSTM, and the Adaptive Ensemble. The ensemble therefore does not dominate at the shortest horizon. This is important because it rules out an interpretation in which combination automatically improves every forecasting problem.

At one month, the Adaptive Ensemble becomes the empirical RMSE leader with RMSE 21.00. Its improvement over Historical Median is 3.09%, which is not statistically distinguishable at the 5% level. From two months onward, the relative advantage becomes increasingly pronounced. At 12 months, for example, AE RMSE is 52.38 compared with 100.51 for Historical Median. At 24 and 36 months, the corresponding AE values are 42.08 and 37.12, compared with 119.02 and 97.28 for the best non-ensemble alternatives.

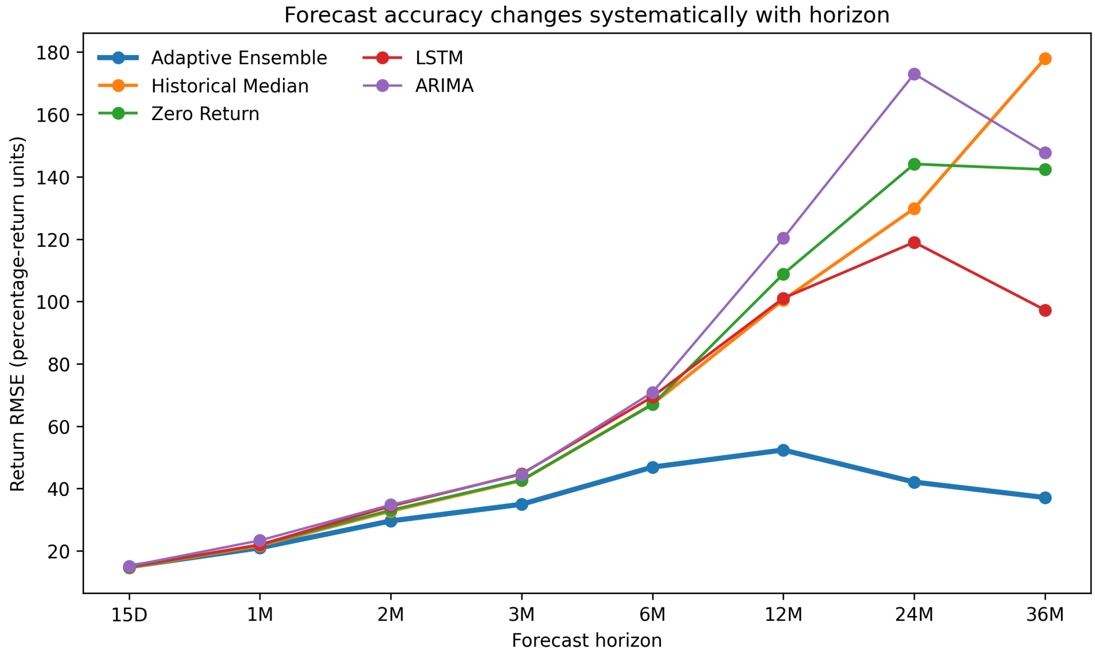

Figure 1: Return RMSE across forecast horizons for the principal forecasting specifications. The figure emphasizes the change in model ranking rather than the absolute scale of the cumulative-return target.

Table 3: Adaptive Ensemble versus the best non-ensemble specification. RMSE and MAE are percentage-return units. The p-value is from the one-sided HAC Diebold–Mariano comparison.

Horizon	AE RMSE	Best non-AE	RMSE	Improvement (%)	AE MAE	p
15D	14.94	Historical Median	14.61	-2.26	10.71	0.738
1M	21.00	Historical Median	21.67	3.09	15.79	0.061
2M	29.62	Historical Median	32.75	9.57	22.29	¡0.001
3M	34.96	Historical Median	42.58	17.90	26.52	¡0.001
6M	46.88	Zero Return	66.95	29.98	33.44	0.002
12M	52.38	Historical Median	100.51	47.89	35.79	0.002
24M	42.08	LSTM	119.02	64.64	29.36	0.037
36M	37.12	LSTM	97.28	61.84	29.40	¡0.001

Table 3 provides the clearest evidence for H2 and H3. The Adaptive Ensemble does not significantly outperform the best non-ensemble benchmark at 15 days or one month at the 5% level. From two months onward, the one-sided HAC Diebold–Mariano test rejects equal predictive accuracy in favor of the Adaptive Ensemble. The result is therefore stronger than a simple point-ranking statement, but it is not evidence of universal dominance.

Figure 2 shows the relative improvement directly. The gain rises from 9.57% at two months to 47.89% at 12 months and remains above 60% at 24 and 36 months. Because the target itself changes with h, the relevant comparison is relative model performance within each horizon rather than monotonicity in the absolute RMSE series.

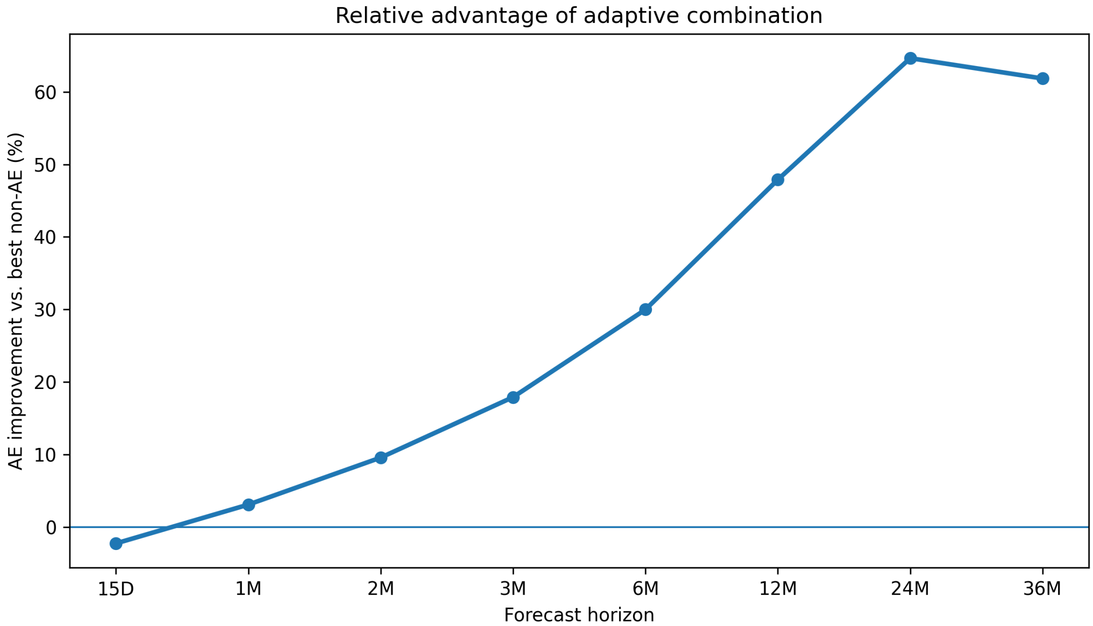

Figure 2: Relative RMSE improvement of the Adaptive Ensemble over the best non-ensemble specification at each horizon. Positive values indicate lower RMSE for the Adaptive Ensemble.

7.2 MAE and magnitude forecasting

The MAE results reinforce the broad RMSE pattern. AE MAE values are 10.71, 15.79, 22.29, 26.52, 33.44, 35.79, 29.36, and 29.40 from 15 days through 36 months. The evidence therefore does not depend on a single squared-error statistic.

7.3 Directional performance and the base-rate problem

Directional results reveal a different pattern. The Adaptive Ensemble has 58.23% directional accuracy at 15 days and 58.77% at one month. Accuracy rises to 65.27% at two months, 75.10% at six months, and 92.05% at 12 months. It reaches 99.81% at 36 months.

Table 4: Directional performance of the Adaptive Ensemble. DA = directional accuracy; BA = balanced accuracy; MCC = Matthews correlation coefficient; Excess = DA minus the Always-Up accuracy, in percentage points.

Horizon	DA (%)	BA (%)	MCC	Always-Up (%)	Excess (pp)
15D	58.23	53.77	0.121	55.73	2.49
1M	58.77	53.84	0.120	56.44	2.32
2M	65.27	58.68	0.296	58.37	6.89
3M	64.59	56.83	0.287	59.03	5.56
6M	75.10	65.32	0.470	64.11	10.99
12M	92.05	83.70	0.780	75.69	16.36
24M	87.88	75.44	0.662	75.33	12.55
36M	99.81	70.00	0.632	99.68	0.13

The 36-month observation deserves special emphasis. The 99.81% directional accuracy is extraordinary in isolation, but 99.68% of realized returns are positive. An Always-Up rule therefore obtains almost the same accuracy. The Adaptive Ensemble's excess is only 0.127 percentage points, and the exact McNemar test does not reject the null at the 5% level ($p = 0.50$). This non-rejection does not establish equivalence; it means only that the paired classifications are not statistically distinguishable under the exact McNemar test. H5 is therefore strongly supported: near-perfect long-horizon DA cannot be interpreted as evidence of near-perfect conditional predictability.

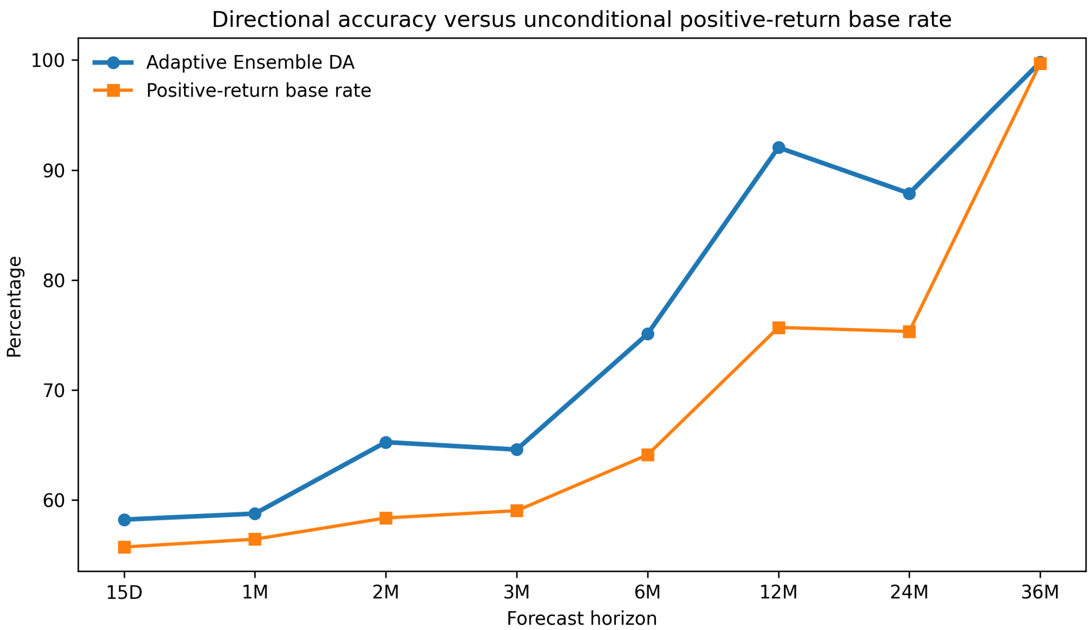

Figure 3: Adaptive Ensemble directional accuracy compared with the unconditional positive-return base rate. The convergence at 36 months illustrates why directional accuracy must be interpreted jointly with the class distribution.

The 12-month result is different. The positive-return rate is 75.69%, while AE reaches 92.05%

accuracy and exceeds Always-Up by 16.36 percentage points. Thus, the 12-month result contains substantially more directional information than the 36-month result. Directional accuracy should consequently be reported with base-rate-adjusted measures rather than in isolation.

7.4 Model-set uncertainty

The block-bootstrap Hansen-style diagnostic retains all candidate models at every horizon and at $\alpha = 0.05$, 0.10, and 0.15. This is not evidence that all models have identical predictive performance. Rather, it indicates that the observed loss differences are not sufficiently large, under this model-set procedure, to isolate a statistically distinguishable superior subset.

This result is complementary to the RMSE ranking. The Adaptive Ensemble can be the empirical RMSE leader while model-set uncertainty remains substantial. Point estimates answer which model has the lowest realized average loss in the sample; the model-set diagnostic asks whether the observed differences are large enough to justify excluding alternatives. The present evidence supports the first statement but not the second.

7.5 Post-hoc ablation of forecast combination

A post-hoc ablation was reconstructed from the stored out-of-sample forecasts. No component model was refitted. Three equal-weight combinations are compared with the Adaptive Ensemble: a classical group, a neural group, and all nine magnitude-forecast components.

Table 5: Post-hoc equal-weight combinations reconstructed from stored out-of-sample forecasts. No model was refitted for this analysis. Values are RMSE in percentage-return units.

Horizon	Adaptive Ensemble	Classical equal-weight	Neural equal-weight	All magnitude equal-weight
15D	14.94	55.38	14.90	40.66
1M	21.00	238.78	22.35	257.12
2M	29.62	362.59	33.86	426.02
3M	34.96	359.31	43.62	437.59
6M	46.88	64.57	67.82	61.56
12M	52.38	83.81	100.53	82.67
24M	42.08	101.09	119.76	98.94
36M	37.12	70.36	100.94	64.27

The adaptive rule improves on the equal-weight neural combination from one month onward and by increasingly large margins at medium and long horizons. Equal weighting across all magnitude forecasts is particularly unstable at one to three months because extreme GLS-type forecasts receive the same weight as stable benchmarks. The purpose of this ablation is not to prove that adaptive weighting is universally optimal; rather, it demonstrates that the observed gains are not simply a consequence of averaging more forecasts.

7.6 Adaptive weights and the mechanism of combination

The ensemble weights are not fixed. Figure 4 and Table 6 show the average weights across forecast origins. The early part of each horizon necessarily begins with equal weights because a 20-error history is required. Thereafter, the weights respond to recent forecast performance. In particular, unstable GLS and GLS-LSTM forecasts receive lower average influence than they would under equal weighting in several horizons.

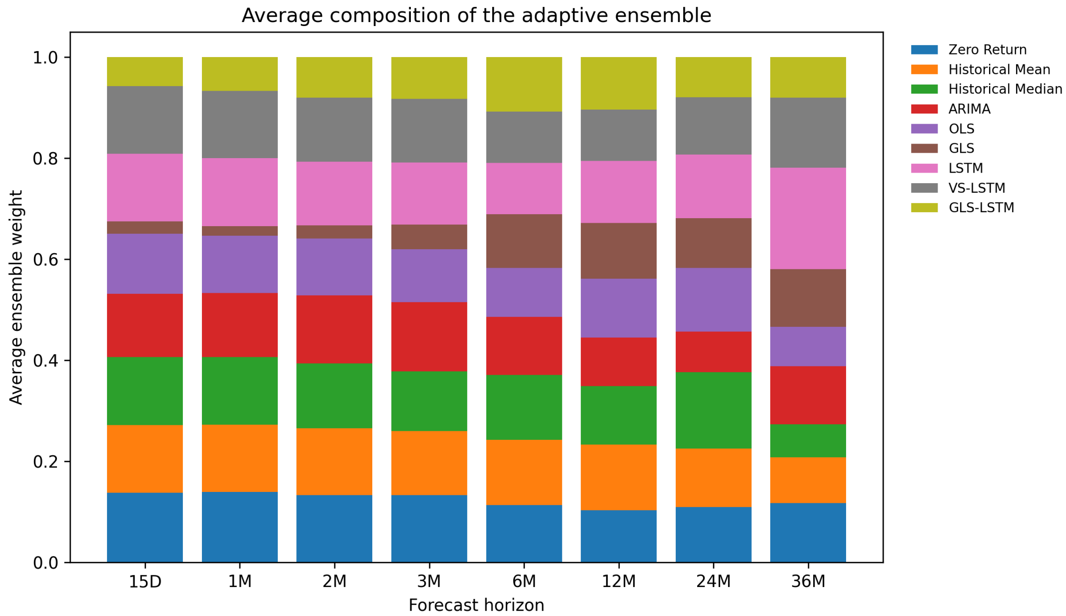

Figure 4: Average composition of the Adaptive Ensemble across forecast horizons. Weights are computed causally from the preceding 20 out-of-sample errors.

Table 6: Average Adaptive Ensemble weights by horizon.

Horizon	Zero	Mean	Median	ARIMA	OLS	GLS	LSTM	VS-LSTM	GLS-LSTM
15D	0.138	0.134	0.135	0.125	0.120	0.024	0.134	0.134	0.057
1M	0.140	0.133	0.134	0.127	0.113	0.019	0.135	0.133	0.067
2M	0.134	0.132	0.129	0.134	0.113	0.026	0.126	0.127	0.080
3M	0.133	0.127	0.118	0.137	0.104	0.049	0.123	0.126	0.082
6M	0.113	0.129	0.128	0.115	0.097	0.106	0.102	0.101	0.108
12M	0.103	0.130	0.116	0.096	0.117	0.110	0.123	0.102	0.103
24M	0.109	0.116	0.151	0.080	0.126	0.099	0.126	0.114	0.079
36M	0.117	0.091	0.065	0.115	0.078	0.114	0.201	0.139	0.080

The weighting mechanism therefore provides a plausible explanation for why the ensemble can remain competitive despite the presence of highly unstable component forecasts. It does not remove those forecasts from the candidate set; instead, it reduces their influence when their recent errors are large. This is precisely the behavior expected from a recency-adaptive combination rule.

7.7 Recent-sample and overlap robustness

The recent-half RMSE values are compared with full-sample RMSE in Figure 5 and Table 7. The medium- and long-horizon advantage remains visible when attention is restricted to the latter half of the evaluation period.

Table 7: Full-sample and recent-half RMSE for the Adaptive Ensemble.

Horizon	Full sample	Recent half
15D	14.94	13.58
1M	21.00	19.03
2M	29.62	26.96
3M	34.96	30.41
6M	46.88	44.35
12M	52.38	42.39
24M	42.08	30.16
36M	37.12	33.24

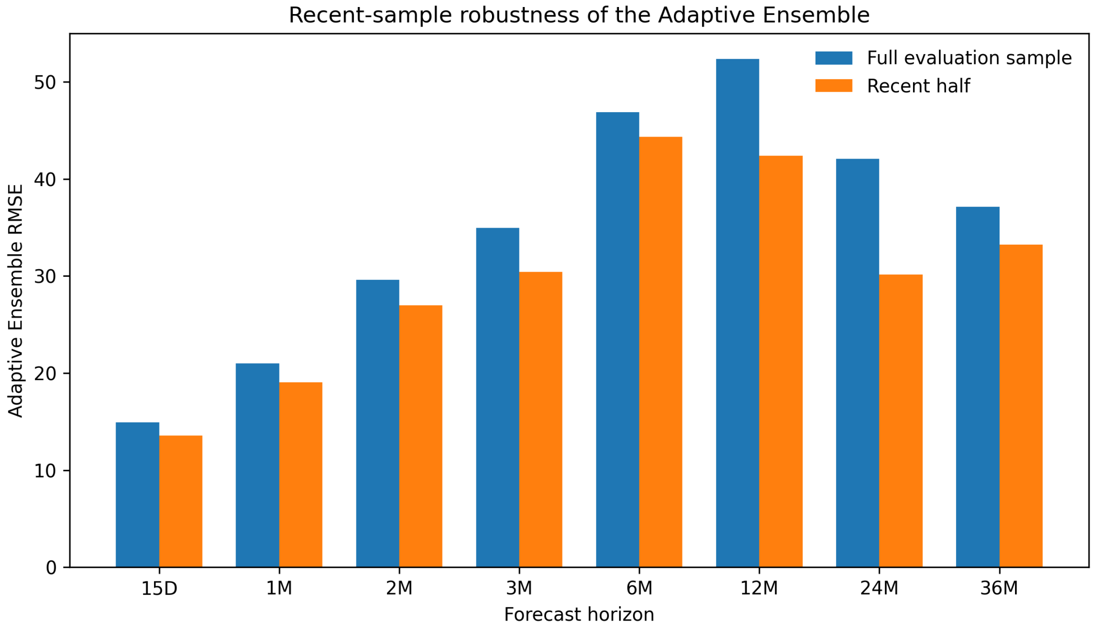

Figure 5: Adaptive Ensemble RMSE in the full evaluation sample and the recent half of the evaluation sample.

The non-overlapping analysis is much weaker as an inferential device. At 12, 24, and 36 months, the effective non-overlap samples are approximately 9, 4, and 2 observations, respectively. These sample sizes are insufficient for reliable standalone model ranking. The non-overlap exercise is therefore treated only as a diagnostic rather than as confirmatory evidence.

7.8 Forecast-error stability and target diagnostics

CUSUM diagnostics reject mean stability of Adaptive Ensemble forecast errors across all horizons. The result indicates that average forecast errors are not temporally constant. It does not by itself prove a particular structural-break mechanism or establish that adaptive weighting is optimal. Rather, it provides a rationale for considering adaptive rather than exclusively fixed-weight combination schemes.

The stationarity diagnostics also change with horizon. Short-horizon targets are broadly compatible with stationary behavior, whereas the 12–36 month targets show mixed or nonstationary indications. Because direct cumulative-return targets overlap by construction, these tests are interpreted descriptively rather than as standalone evidence for a specific data-generating process.

8 Discussion

The central empirical result is a clear horizon dependence in Bitcoin forecasting performance. At 15 days, Historical Median outperforms the Adaptive Ensemble by a small amount. At one month, the ensemble becomes the empirical RMSE leader, but the 3.09% improvement is not statistically distinguishable at the 5% level. From two months onward, the advantage becomes both economically visible in the error metrics and statistically distinguishable under HAC Diebold–Mariano inference. H1 and H2 are therefore supported in the sense that model rankings change with horizon and adaptive combination performs particularly well at medium and long horizons.

The result is consistent with the broader forecast-combination literature. Combination can be valuable when component forecasts contain complementary information and their errors are not perfectly aligned (Bates and Granger, 1969; Timmermann, 2006). At short horizons, simple distributional features can be difficult to beat. As the horizon changes, heterogeneous model errors can become more consequential, making adaptive weighting useful. This interpretation is deliberately phrased as a mechanism consistent with the evidence, not as a structural proof about Bitcoin's data-generating process.

The findings also reinforce the point that model complexity should not be equated with forecasting superiority. LSTM is competitive at several horizons, while VS-LSTM does not establish a universal advantage. The result is consistent with recent cryptocurrency forecasting studies that report heterogeneous rankings across model classes and information sets (Berger and Koubová, 2024; AlMadany et al., 2024; Bouteska et al., 2024; Gurgul et al., 2025). The practical implication is that algorithmic comparison should be conducted within a common out-of-sample framework rather than interpreted as a permanent hierarchy.

The Adaptive Ensemble is also not a black-box superlearner. Its weighting rule is transparent and based on recent out-of-sample MAE. This creates a compromise between fixed equal weighting and fully optimized combination weights. The post-hoc ablation indicates that this particular adaptive rule is more stable than several equal-weight constructions in the present sample. The evidence should nevertheless not be generalized to all adaptive weighting rules.

The model-set result provides an important qualification. The Adaptive Ensemble is the lowest-RMSE specification from one month onward, but the block-bootstrap model-set diagnostic retains all candidate models. Thus, point estimates and statistical model selection answer different questions. The point estimates identify the model with the lowest realized average loss; the model-set diagnostic asks whether the differences are sufficiently large to exclude alternatives. The present evidence supports the former but not the latter.

The directional results provide an even stronger methodological lesson. Long-horizon directional accuracy can become deceptively high when the unconditional sign distribution is strongly asymmetric. At 36 months, 99.68% of realized returns are positive. The Adaptive Ensemble's 99.81% directional accuracy is therefore almost entirely compatible with an Always-Up rule. Its excess accuracy is only 0.127 percentage points and is not statistically distinguishable under the exact McNemar test. The result should not be described as evidence of near-perfect conditional forecasting.

The 12-month result illustrates why the qualification is not merely a criticism of directional metrics. At that horizon, the positive-return base rate is 75.69%, whereas AE achieves 92.05% accuracy and an excess of 16.36 percentage points over Always-Up. This difference suggests that base-rate adjustment can materially change the interpretation of directional results.

The stability diagnostics further suggest that forecasting relationships evolve over time. CUSUM evidence of forecast-error instability indicates that average predictive errors are not temporally constant. This supports evaluating adaptive procedures rather than assuming that a fixed model ranking should remain permanent. It does not, however, establish that the particular 20-error weighting window used here is optimal.

Several methodological limitations should be explicit. First, the target is a cumulative return and overlapping observations induce dependence. The full overlapping sample is retained to preserve statistical power, while HAC inference and robustness diagnostics address the dependence. Second, non-overlapping long-horizon samples are extremely small. Third, the adaptive weights use a fixed 20-error history and models are retrained every 25 origins; neither choice is claimed to be optimal. Fourth, the model-set procedure is a Hansen-style bootstrap approximation rather than an exact implementation of the canonical MCS algorithm. Fifth, extreme forecasts from some linear specifications, particularly GLS and GLS-LSTM, can strongly affect equal-weight combinations; this is why the ablation is interpreted as a diagnostic of combination stability rather than as a universal comparison of ensemble classes.

Finally, the study does not establish economic profitability. Lower RMSE does not automatically imply a profitable trading strategy after transaction costs, liquidity constraints, execution delays, turnover, or risk adjustment. Nor does statistical directional accuracy establish exploitable conditional expected returns. The contribution is narrower and more defensible: under a rigorous out-of-sample forecasting design, the relative predictive performance of Bitcoin models changes with horizon, and adaptive forecast combination becomes increasingly useful at medium and long horizons.

9 Conclusion

This study examined whether Bitcoin forecasting performance changes with forecast horizon and whether adaptive combination can exploit differences among heterogeneous forecasting models. The evidence supports a clear horizon-dependent interpretation. Historical Median is the best RMSE specification at 15 days, while the Adaptive Ensemble becomes the empirical RMSE leader from one month onward.

The strongest inferential evidence appears from two months onward. Relative to the best non-ensemble specification, the Adaptive Ensemble reduces RMSE by 9.57% at two months, 17.90% at three months, 29.98% at six months, 47.89% at 12 months, 64.64% at 24 months, and 61.84% at 36 months. These differences are statistically distinguishable under one-sided HAC Diebold–Mariano testing from two months onward. The one-month advantage is smaller and does not reach the conventional 5% threshold.

The model-set evidence imposes an important qualification. The block-bootstrap Hansen-style diagnostic does not isolate a unique superior model. The appropriate conclusion is therefore not that the Adaptive Ensemble is universally optimal, but that adaptive combination is a useful horizon-dependent forecasting strategy under the present data, model set, and evaluation design.

The directional results provide a second qualification. At 36 months, the Adaptive Ensemble reaches 99.81% directional accuracy, but 99.68% of realized returns are positive. The excess over an Always-Up rule is only 0.13 percentage points and is not statistically distinguishable. Long-horizon directional accuracy must therefore be interpreted against the unconditional sign distribution.

The broader implication is methodological: Bitcoin forecasting should be evaluated as a collection of horizon-specific problems rather than as a single competition among algorithms. Future research should examine alternative adaptive weighting rules, probabilistic and quantile forecasts, additional real-time data vintages, richer high-dimensional information sets, and economic evaluation incorporating transaction costs, turnover, drawdown, and risk-adjusted performance. Transformer and graph-based architectures may be useful additions, but they should be evaluated under the same leakage-aware walk-forward framework.

Data and Computational Transparency

The analysis is based on a frozen daily modeling dataset and an auditable forecast panel. Each forecast record contains an origin, horizon, target date, actual return, predicted return, and, for ensemble forecasts, the weights available at that origin. The computational design fixes model specifications and random seeds, records the retraining frequency, and permits independent reconstruction of the primary RMSE, MAE, and directional metrics from the stored forecast panel. The post-hoc ablation uses those stored out-of-sample forecasts and does not refit the component models.

Declaration of Competing Interest

The author declares no known competing financial or personal interests that could have appeared to influence the work reported in this paper.

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Data Availability

The frozen modeling dataset used in this study is publicly available as part of the replication repository at:

`https://github.com/moradimojtaba/bitcoin-forecasting/blob/main/data/processed/bitcoin_modeling_dataset.csv`

The dataset spans 14 May 2011 to 25 May 2024 and contains daily Bitcoin prices, blockchain on-chain variables, and derived features. The underlying third-party source data remain subject to the terms of the respective data providers; the processed dataset is distributed under the same license as the replication materials (MIT License).

Code Availability

The complete analysis code is publicly available at:

`https://github.com/moradimojtaba/bitcoin-forecasting`

The main Python script is located at:

`https://github.com/moradimojtaba/bitcoin-forecasting/blob/main/code/bitcoin_forecasting.py`

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